5.7 Inverse of a Function

Name _____ Period ____ Date _____

Algebra II 5.7 · Student sheet

I can…

- find the inverse of a linear function.
- find the inverse of a nonlinear function, restricting the domain when needed.
- verify that two functions are inverses.

Vocabulary · fill in the missing word

Inverse functions — functions that _____ each other; f^{-1} reverses the inputs and outputs of f

Finding an inverse — _____ x and y, then solve for y

Horizontal line test — the inverse of f is a _____ when no horizontal line crosses the graph of f more than once

Verifying inverses — f(g(x)) = _____ = x

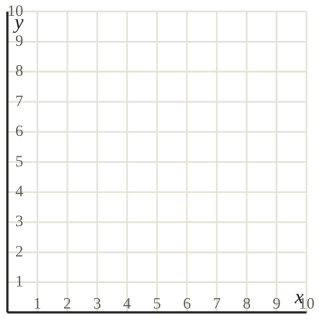
Part 1 · Undo it: temperature

F = 1.8C + 32 changes degrees Celsius to degrees Fahrenheit. Complete the table, then swap the columns.

C	0	10	20	30
F				

F (swapped)				
C	0	10	20	30

Graph both sets of points and the line y = x. (Use 10 units per box.)



What do you notice? _____

Solve F = 1.8C + 32 for C: C = _____

Part 2 · Notes: fill in the blanks

To find an inverse: replace f(x) with y, _____ x and y, then solve for _____.

The graphs of f and f^{-1} are reflections in the line _____.

Horizontal line test: the inverse of f is a function when no horizontal line crosses the graph of f more than _____.

To verify that f and g are inverses, show that _____ = x and _____ = x.

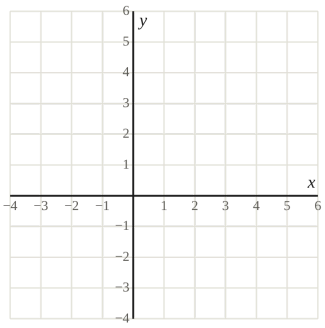
Part 3 · Examples (work them with the class)

Example 1

Let $f(x) = 2x + 3$. Solve $y = f(x)$ for x. Then find the input when the output is -7.

Example 2

Find the inverse of $f(x) = 3x - 1$. Then graph f and f^{-1}.

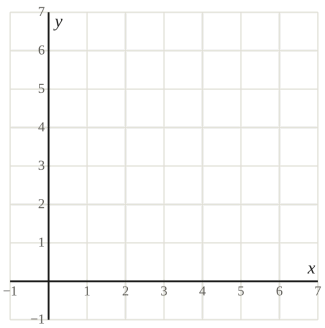


Example 3

Find the inverse of $f(x) = 2x^3 + 1$.

Example 4

Find the inverse of $f(x) = x^2 + 2$, $x \geq 0$. Why is the restriction needed?



Example 5

Verify that $f(x) = 3x - 1$ and $g(x) = \dfrac{x + 1}{3}$ are inverse functions.

Example 6

The formula $F = 1.8C + 32$ converts degrees Celsius to degrees Fahrenheit. Find the inverse and use it to convert 77°F to Celsius.

Part 4 · Partner practice

Partner 1

Find the inverse of $f(x) = -4x + 8$.

Partner 2

Find the inverse of $f(x) = \sqrt{x - 3}$.